

Ruidong Zhang

M.S. Student in Computer Science

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Summary

Master's student at the University of Science and Technology of China working on compiler systems, GPU and AI systems, and hardware-software co-design.

Education

M.S.	University of Science and Technology of China (USTC) , Computer Science	Hefei, China Sept 2025 – present
B.S.	University of Electronic Science and Technology of China (UESTC) , Computer Science • GPA: 3.84 / 4.00	Chengdu, China Sept 2021 – June 2025

Research Experience

Shanghai Artificial Intelligence Laboratory , Research Intern — High-Performance Compilation	Shanghai, China Jan 2026 – July 2026
• Worked on MLIR-based compiler infrastructure and performance optimization for GPU-intensive workloads, covering compiler IRs, transformations, lowering, and GPU code generation. • Implemented and evaluated compiler transformations related to program rewriting and memory-system optimization. • Profiled generated CUDA workloads across GPU platforms using low-level performance analysis, studying memory traffic, synchronization, parallelism, and occupancy bottlenecks.	

Industry Experience

Ubiquant Technology , Quantitative Implementation Intern — LLM Systems	Shanghai, China Aug 2026 – Sept 2026
• Evaluated multi-node, multi-GPU LLM serving with SGLang and vLLM under prefill-decode disaggregation, characterizing TTFT/TPOT trade-offs across distributed execution configurations. • Investigated Mooncake-based KV-cache transfer and profiled compute, memory, and communication bottlenecks across prefill/decode placement strategies.	
Houmo Technology , AI Compiler Development Intern	Nanjing, China Apr 2025 – July 2025
• Implemented 44 ONNX operators end-to-end for a RISC-V-based compute-in-memory NPU, spanning MLIR dialect definitions, pattern rewriting, and lowering to accelerator-specific primitives. • Built MLIR lowering and validation pipelines that mapped ONNX operators to backend execution constraints and regression-tested outputs against ONNX Runtime.	

Projects

Compiler Techniques for Stateful Simulation — Ongoing	June 2026 – present
Exploring compiler techniques for repeated stateful simulation workloads, with an emphasis on identifying reusable computation while preserving numerical-solver and runtime semantics.	
RISC-V Processor Design & Architectural Validation	Apr 2024 – Oct 2025
Designed a five-stage RV32E/RV32I processor and deployed it on an Artix-7 FPGA to boot RT-Thread. • Added forwarding and hazard control, branch handling, CSR/exception/interrupt support, and an AXI4 memory subsystem. • Built an RV32IM emulator with breakpoints, disassembly, and device simulation, together with Spike differential testing for ISA-level validation.	

Technical Skills

Programming: C++, C, Python, CUDA, Verilog
Compilers: MLIR, LLVM
GPU & AI Systems: CUDA, ONNX, PyTorch, SGLang, vLLM

Selected Honors

- IEEEExtreme Programming Competition — Top 5% Worldwide
- HP Responsible Entrepreneurship Competition — First Prize